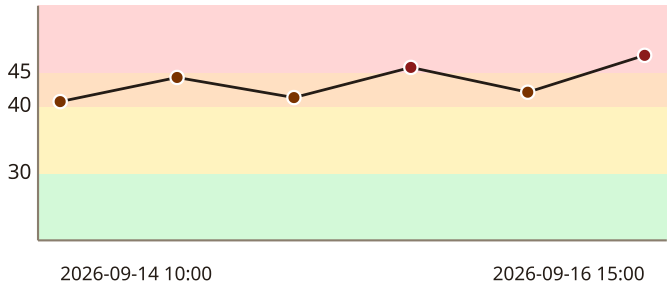
# Hot Room Log

Sample data — fictional rooms and readings for demonstration.

## Heat evidence sheet: Demo Estate Study Room

Sample

- Room: Demo Estate Study Room
- Readings: 6
- Latest: 47.6 (Dangerous) on 2026-09-16 15:00
- Period: 2026-09-14 to 2026-09-16
- Highest: 47.6 (Dangerous) on 2026-09-16 15:00
- Great discomfort or above: 6 of 6




| Time | Temperature (°C) | Humidity (%) | Dew point (°C) | Humidex | Band | Note |
|---|---|---|---|---|---|---|
| 2026-09-14 10:00 | 29 | 78 | 24.8 | 40.7 | Great discomfort | |
| 2026-09-14 15:00 | 32 | 68 | 25.3 | 44.3 | Great discomfort | |
| 2026-09-15 10:00 | 29.5 | 76 | 24.8 | 41.3 | Great discomfort | |
| 2026-09-15 15:00 | 33 | 66 | 25.8 | 45.8 | Dangerous | Fan on, windows open |
| 2026-09-16 10:00 | 30 | 75 | 25.1 | 42.1 | Great discomfort | |
| 2026-09-16 15:00 | 34 | 65 | 26.5 | 47.6 | Dangerous | |

### How these numbers were calculated

```
es = 6.105 × exp(17.27 × T ÷ (237.7 + T))
```

```
e = RH ÷ 100 × es
```

```
humidex = T + 0.5555 × (e − 10)
```

```
gamma = ln(RH ÷ 100) + 17.27 × T ÷ (237.7 + T)
```

```
dew point = 237.7 × gamma ÷ (17.27 − gamma)
```

Comfortable is below 30. Some discomfort is 30 to 39.9. Great discomfort is 40 to 45. Dangerous is above 45.

Humidex and dew point are rounded half away from zero to one decimal place. The humidity effect is the humidex minus the air temperature, also to one decimal place.

Readings were typed in by hand; accuracy depends on the instrument used.

Dear Room manager,

We are students who use Demo Estate Study Room. From 2026-09-14 to 2026-09-16, we recorded 6 temperature and humidity readings in this room.

The highest humidex was 47.6 (Dangerous) on 2026-09-16 15:00, when the air temperature was 34 °C and the relative humidity was 65 %. Humidity added 13.6 humidex points on top of the air temperature.

Readings by humidex comfort band: Dangerous 2, Great discomfort 4.

Could you please check the ventilation, fans or air-conditioning in this room? We can share the full list of readings.

Thank you,
[your name]

Generated by Hot Room Log on 2026-10-11.

Humidex bands are a comfort guide, not medical advice.

Hot Room Log · solo project by Zhenhua Liu for STEMergent Hacks 2026